

MAT31 - 6

Math for Data Practice

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About this textbook

This year, for the first time, every Bachelor course at Albert School has its own textbook, thanks to your teachers and AI.

- The textbook is a help for working on the course at home, after the class.
- The course as taught in class remains the reference, and prevails over the textbook wherever the two differ.
- This is a first version. Help us improve it by sharing your feedback with your teacher.

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Preface

By the end of this course you will be able to do six things you cannot do yet. You will measure how strongly two quantities move together, and know exactly what that measurement does and does not capture. You will fit the least-squares line and read its slope as a business quantity. You will say how much of the variation in the data the line accounts for, through the number R^2 . You will check, from the residuals, whether the assumptions the line rests on actually hold. You will test whether the slope could be an accident of noise. And you will recognise when a hidden third variable has reversed the story completely. Together these are one skill: fitting a straight line to a cloud of points, and then deciding how much to trust it. Any cloud of points admits a line; the task is to tell a line that means something from a line that means nothing.

No chapter teaches the tools you arrive with, and you need every one of them. From your earlier statistics you can summarise one variable by its mean and its variance, and you have met two ways to relate a second variable to it: the covariance and Pearson's correlation, read alongside a scatter plot. From multivariable calculus you can find the lowest point of a function of two variables by setting its two partial derivatives to zero. From probability and inference you know random variables, their expectation and variance, the idea of an estimator, the normal distribution, the Student t distribution, and how a hypothesis test turns a test statistic into a p -value.

We work throughout with one small dataset — thirty regional markets of an online retailer, each with a week's advertising spend and the sales that followed — and we compute everything from the ground up in Python with NumPy, calling no regression library. Where a statistics package would print a summary table in one line, we form the slope, its standard error, and the test statistic ourselves, so that every number is one you could have produced by hand. Every code block on these pages has been run, and the output printed beneath it is its real output.

1. Covariance and correlation

Before any line is drawn, we need a single number for how strongly two quantities move together. A retailer looking at thirty markets wants to know: where advertising spend is high, are sales high too — and how tightly? That number is the correlation, and it is built from a more basic one, the covariance.

Throughout the chapter the data are n pairs $(x_1, y_1), \dots, (x_n, y_n)$. The index i runs over the markets, from 1 to n ; for market i , the first coordinate x_i is the advertising spend and the second coordinate y_i is the sales, both measured in thousands of euros. We write $\bar{x}$ and $\bar{y}$ for the two means.

1.1. From variance to covariance

For one variable, the variance measures spread: the average squared distance of each value from its mean. With two variables we ask instead whether they stray from their means together. For each market, multiply the two deviations $x_i - \bar{x}$ and $y_i - \bar{y}$: this product is positive when both coordinates land on the same side of their means (both above, or both below) and negative when they land on opposite sides. Adding these products and dividing gives the covariance.

Definition 1 (Sample covariance). For paired data $(x_1, y_1), \dots, (x_n, y_n)$ with means $\bar{x}$ and $\bar{y}$, the **sample covariance** of x and y is the number

$$\text{cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

The covariance is positive when the two variables tend to rise together, negative when one tends to rise as the other falls, and zero when they show no linear co-movement. Taking $y = x$ recovers the sample variance s_x^2 .

The divisor is $n - 1$, not n , for the same reason it is in the sample variance. The deviations are taken about the sample means $\bar{x}$ and $\bar{y}$, which are themselves computed from the data; the n deviations $x_i - \bar{x}$ therefore already satisfy one constraint, namely that they sum to zero, so only $n - 1$ of them can be chosen freely. Dividing by that number of free deviations, rather than by n , corrects a systematic underestimate: the population covariance measures deviation about the true, unknown means, and deviation about the sample means is always a little smaller. With $n - 1$ the estimate is unbiased. This is the divisor NumPy uses when we pass `ddof=1`, and it is the population covariance, computed about known means with a divisor n , that it does not use.

Example — the sign of the covariance. If larger apartments also tend to cost more, then area and price have positive covariance. If busier roads tend to have cheaper houses, then traffic and price have negative covariance. The sign reports the direction of the linear trend. The size, on its own, is hard to read — which is the problem correlation solves.

1.2. The problem with covariance: units

Covariance has one flaw as a summary: its size depends on the units. Measure spend in euros instead of thousands of euros and every deviation $x_i - \bar{x}$ becomes a thousand times larger, so the covariance

becomes a thousand times larger — yet nothing about the relationship has changed. A number that changes when you switch from metres to centimetres cannot, by itself, tell you whether a relationship is strong.

The remedy is to divide by the spread of each variable, which cancels both sets of units. The result is Pearson's correlation coefficient.

Definition 2 (Pearson's correlation coefficient r). For paired data with sample standard deviations $s_x > 0$ and $s_y > 0$, **Pearson's correlation coefficient** is

$$r = \frac{\text{cov}(x, y)}{s_x s_y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}.$$

Pearson's r is a pure number, free of units, and it always lies in the interval $[-1, 1]$, as the proposition below proves. The two forms are the same number: in the second, the $n - 1$ divisors of the covariance and of the two variances cancel, leaving only raw sums of squares. It is convenient to name those sums, as they return in every chapter that follows.

Definition 3 (Sums of squares and cross-products). For paired data, the three **sums of squares and cross-products** are

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2, \quad S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2, \quad S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

In this notation, $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$. The bound on r is worth proving, because the proof also explains what r means geometrically.

Proposition 1 (r lies between -1 and 1). For any paired data with $s_x > 0$ and $s_y > 0$, one has $-1 \leq r \leq 1$. The value $r = \pm 1$ occurs exactly when the points lie on a straight line of nonzero slope.

Proof. Collect the two lists of deviations into vectors of $\mathbb{R}^n$, namely $u = (x_i - \bar{x})_{1 \leq i \leq n}$ and $v = (y_i - \bar{y})_{1 \leq i \leq n}$. Then $S_{xy} = \langle u, v \rangle$ is their inner product, $S_{xx} = \|u\|^2$ and $S_{yy} = \|v\|^2$, so

$$r = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

is the cosine of the angle between u and v . The Cauchy-Schwarz inequality states that $|\langle u, v \rangle| \leq \|u\| \|v\|$, which gives $|r| \leq 1$. Cauchy-Schwarz is an equality exactly when u and v are collinear, that is when $y_i - \bar{y}$ is one fixed multiple of $x_i - \bar{x}$ for every i ; this is precisely the statement that the points (x_i, y_i) lie on a straight line of nonzero slope. $\square$

So r is the cosine of the angle between the two centred data vectors: $r = 1$ means they point the same way, a perfectly rising line; $r = -1$ means they point opposite ways, a perfectly falling line; and $r = 0$ means they are perpendicular, with no linear alignment at all.

1.3. Computing them on the markets

We load the thirty markets and compute both quantities. NumPy returns the covariance as one entry of a small matrix, and the correlation the same way.

```
import numpy as np

# Thirty regional markets of an online retailer. For each market i the record is
# one week's advertising spend x_i and the sales y_i that followed, both in
# thousands of euros (k€). The spend is what the retailer sets; the sales are
# what it hopes the spend drives.
x = np.array([2.1, 2.9, 3.3, 3.3, 3.5, 3.6, 3.8, 4.2, 4.4, 6.9,
              7.0, 7.3, 7.4, 7.9, 9.4, 10.1, 10.3, 10.4, 10.4, 11.4,
              11.9, 13.0, 13.1, 13.5, 14.5, 14.6, 16.5, 16.6, 18.5, 19.8])
y = np.array([3.5, 10.0, 7.4, 11.4, 13.3, 10.7, 5.2, 13.3, 12.5, 19.0,
              19.8, 24.4, 22.6, 16.5, 21.5, 26.9, 25.1, 21.2, 21.6, 21.5,
              25.3, 27.3, 31.2, 25.4, 23.7, 28.7, 29.5, 39.7, 44.6, 40.2])
n = len(x)

print("number of markets:", n)
print("mean spend x-bar :", round(np.mean(x), 2), "k€")
print("mean sales y-bar :", round(np.mean(y), 2), "k€")
```

```
number of markets: 30
mean spend x-bar : 9.39 k€
mean sales y-bar : 21.43 k€
```

```
import numpy as np

# The covariance is the off-diagonal entry of the 2x2 covariance matrix; ddof=1
# selects the n-1 divisor. Pearson's r is the matching off-diagonal entry of the
# correlation matrix -- already unit-free and already inside [-1, 1].
cov_xy = np.cov(x, y, ddof=1)[0, 1]
r = np.corrcoef(x, y)[0, 1]

print("cov(x, y):", round(cov_xy, 2))
print("r          :", round(r, 4))
```

```
cov(x, y): 47.71
r          : 0.9342
```

The correlation is $r = 0.9342$ — strong and positive. Spend and sales rise together tightly. That single number is the summary the retailer asked for, and the starting point for everything that follows.

1.4. What correlation does not capture

Here is the trap. Pearson's r measures linear alignment only. A relationship can be exact and still have r close to zero, if its shape is curved. Figure 1 shows three clouds: a strong rising trend, a strong falling trend, and a clean parabola whose r is essentially zero even though y is almost perfectly determined by x .

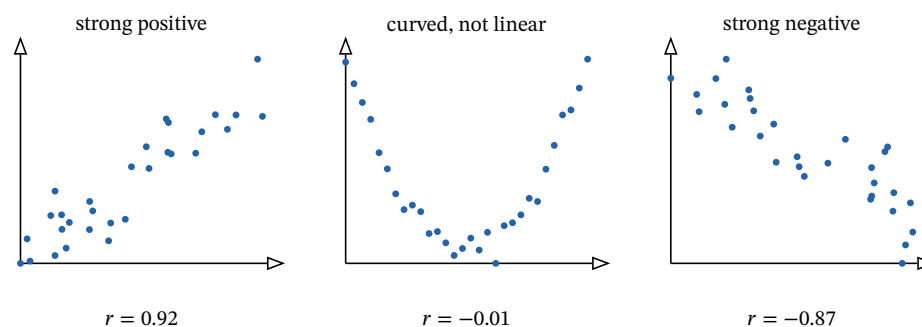


Figure 1: Pearson's r measures only the linear part of a relationship. Left: a strong rising trend, with r near 0.9. Middle: a parabola, where y is almost fully determined by x , yet r is near 0, because there is no straight-line alignment. Right: a strong falling trend, with r near -0.9 . A correlation near zero rules out a straight line, not a relationship.

Pitfall. Reading r near 0 as “ x and y are unrelated” is the most common correlation error. It means only that no straight line fits. Always look at the scatter plot: a U-shape, a cycle, or a plateau can hide behind a small r . In the other direction, a large r does not prove that a line is the right model, only that one is not obviously wrong.

A second fragility: because every deviation is squared in S_{xx} and S_{yy} , a single point far from the trend can dominate the whole coefficient. Such a point has a name.

Definition 4 (Outlier). In a scatter plot, an **outlier** is an observation that lies far from the pattern set by the rest of the data — far from the straight-line trend the other points follow, or far from the bulk of the values in x or in y .

The value of r falls sharply when just one of ten well-behaved points is mis-recorded as an outlier.

```
import numpy as np

# r is not robust: a single wild point can dominate it. Ten markets sit almost
# perfectly on a line; then one sales figure is mis-recorded far below the rest.
xc = np.array([1., 2., 3., 4., 5., 6., 7., 8., 9., 10.])
yc = np.array([1.2, 2.1, 2.9, 4.2, 5.1, 5.8, 7.2, 7.9, 9.1, 9.8])
print("r, clean cloud :", round(np.corrcoef(xc, yc)[0, 1], 3))

yc[9] = -6.0 # the tenth market's sales are mis-recorded far below the trend
print("r, one bad point:", round(np.corrcoef(xc, yc)[0, 1], 3))
```

```
r, clean cloud : 0.999
r, one bad point: 0.08
```

One corrupted observation pulls r down from 0.999 to 0.08: the ten points no longer look linear at all, because of a single bad entry. Correlation is not robust — one outlier can create it or destroy it.

1.5. Correlation is not causation

A last, older warning. A strong r between two variables can arise because one drives the other, because both are driven by a third variable, or by pure coincidence in a small sample. The second case has a name, and it is the subject of Chapter 6.

Definition 5 (Confounding variable). A **confounding variable** (or **confounder**) for the pair (x, y) is a third variable that influences both x and y . When a confounder is present but left out of the analysis, part of its effect on y is wrongly attributed to x , so the observed association between x and y need not reflect any direct link between them.

A correlation, however strong, cannot on its own tell these three cases apart. Establishing that x causes y needs more than the data of a single observational sample; Chapter 6 shows how badly the confounding case in particular can mislead.

2. The least-squares line

Correlation says how tightly the points follow a line; it does not say which line. To predict the sales of a market we have not yet seen, we need the line itself — a slope and an intercept. This chapter derives them, from the ground up, as the one line that makes a natural measure of error as small as possible.

2.1. The line and the error to make small

We look for a straight line, written $y = ax + b$, with slope a and intercept b , that passes as close as possible to the cloud of points. For a candidate line and a market i , the vertical gap between the observed sales y_i and the height $ax_i + b$ of the line above x_i measures how far the line misses that point. We want every such gap small at once. Squaring each gap — so that misses above and below the line cannot cancel, and large misses count for more — and adding gives one number to make small.

Definition 6 (Residual sum of squares). For paired data $(x_1, y_1), \dots, (x_n, y_n)$, the **residual sum of squares** of the candidate line $y = ax + b$ is the value at (a, b) of

$$\begin{aligned} \text{RSS} : \mathbb{R}^2 &\longrightarrow \mathbb{R} \\ (a, b) &\longmapsto \sum_{i=1}^n (y_i - ax_i - b)^2 \end{aligned}$$

Definition 7 (Least-squares line). The **least-squares line** is the line $y = \hat{a}x + \hat{b}$ whose slope $\hat{a}$ and intercept $\hat{b}$ make $\text{RSS}(a, b)$ as small as possible.

Pitfall. The gaps we square are vertical gaps in y , not the shortest distances from the points to the line. Regression treats x as known and predicts y , so the error that matters is the error in y . This is also why the line predicting y from x differs from the line predicting x from y : the two make gaps small in different directions.

2.2. Deriving the slope and the intercept

The function RSS is a smooth function of the two variables a and b . By the method you know for such functions, any lowest point sits where both partial derivatives are zero. Solving those two equations pins down a single candidate line.

Theorem 1 (Least-squares estimates). The function $\text{RSS}(a, b)$ has exactly one point where both partial derivatives vanish, and at that point

$$\hat{a} = \frac{S_{xy}}{S_{xx}} = r \frac{s_y}{s_x}, \quad \hat{b} = \bar{y} - \hat{a} \bar{x}.$$

Proof. We work in four steps.

Step 1 — the partial derivative in the intercept. Differentiating in b and setting the result to zero,

$$\frac{\partial \text{RSS}}{\partial b} = -2 \sum_{i=1}^n (y_i - ax_i - b) = 0.$$

Dividing by $-2n$ gives $\bar{y} - a\bar{x} - b = 0$, that is $b = \bar{y} - a\bar{x}$: the line passes through the point of averages $(\bar{x}, \bar{y})$.

Step 2 — the partial derivative in the slope. Differentiating in a and setting the result to zero,

$$\frac{\partial \text{RSS}}{\partial a} = -2 \sum_{i=1}^n x_i (y_i - ax_i - b) = 0, \quad \text{so} \quad \sum_{i=1}^n x_i (y_i - ax_i - b) = 0.$$

Step 3 — substitute the first equation into the second. Replacing b by $\bar{y} - a\bar{x}$ from Step 1,

$$\sum_{i=1}^n x_i ((y_i - \bar{y}) - a(x_i - \bar{x})) = 0.$$

Step 4 — recognise the sums. Because the deviations $y_i - \bar{y}$ and $x_i - \bar{x}$ each sum to zero, subtracting $\bar{x}$ from every factor x_i changes nothing in the last sum: $\sum_i \bar{x}((y_i - \bar{y}) - a(x_i - \bar{x})) = 0$. Replacing x_i by $x_i - \bar{x}$ therefore gives

$$\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) - a \sum_{i=1}^n (x_i - \bar{x})^2 = S_{xy} - aS_{xx} = 0,$$

hence $\hat{a} = S_{xy}/S_{xx}$, and then $\hat{b} = \bar{y} - \hat{a}\bar{x}$ from Step 1. This candidate is unique, since each step forced its value. Dividing numerator and denominator of $\hat{a}$ by $n-1$ rewrites it as $\text{cov}(x, y)/s_x^2 = r s_y/s_x$. $\square$

The four-step derivation finds the only candidate. It remains to confirm that the candidate really is the lowest point of RSS, and not some other kind of flat spot. This we prove separately, without the two-variable second-derivative test, by making the error small one variable at a time.

Proposition 2 (The estimates give the global minimum). For data whose x_i are not all equal, $\text{RSS}(a, b) \geq \text{RSS}(\hat{a}, \hat{b})$ for every a and b : the least-squares line has slope $\hat{a}$ and intercept $\hat{b}$.

Proof. We work in two steps.

Step 1 — make the error small in the intercept, at a fixed slope. Fix any slope a . As a function of b alone, $\text{RSS}(a, b) = \sum_i ((y_i - ax_i) - b)^2$ is the sum of squared deviations of the numbers $y_i - ax_i$ about b . That sum is smallest when b is their mean, $\bar{y} - a\bar{x}$. Substituting this best intercept, the smallest error attainable at slope a is the value at a of the function

$$\begin{aligned} g : \mathbb{R} &\longrightarrow \mathbb{R} \\ a &\longmapsto \sum_{i=1}^n ((y_i - \bar{y}) - a(x_i - \bar{x}))^2 \end{aligned}$$

Step 2 — make the resulting one-variable function small in the slope. Expanding,

$$g(a) = S_{yy} - 2aS_{xy} + a^2S_{xx}.$$

Since the x_i are not all equal, $S_{xx} > 0$, so g is an upward parabola in a ; it has one lowest point, where $g'(a) = -2S_{xy} + 2aS_{xx} = 0$, that is at $a = S_{xy}/S_{xx} = \hat{a}$. The best intercept there is $\bar{y} - \hat{a}\bar{x} = \hat{b}$. For any (a, b) we have $\text{RSS}(a, b) \geq g(a) \geq g(\hat{a}) = \text{RSS}(\hat{a}, \hat{b})$, so $(\hat{a}, \hat{b})$ is the global minimum. $\square$

The slope has two readings. As S_{xy}/S_{xx} it is the covariance over the variance of x ; as $r s_y/s_x$ it is the correlation rescaled by how many y -units go with one x -unit. Both use only the five summary numbers $\bar{x}, \bar{y}, s_x, s_y$ and r — no matrices, no iteration.

Example — a line by hand. Take five markets with spend $x = (1, 2, 3, 4, 5)$ and sales $y = (1, 3, 4, 5, 7)$. Then $\bar{x} = 3$ and $\bar{y} = 4$, and

$$S_{xy} = (-2)(-3) + (-1)(-1) + 0 + (1)(1) + (2)(3) = 14, \quad S_{xx} = 4 + 1 + 0 + 1 + 4 = 10.$$

So $\hat{a} = 14/10 = 1.4$ and $\hat{b} = 4 - 1.4 \times 3 = -0.2$: the line is $\hat{y} = 1.4x - 0.2$. It predicts $(1.2, 2.6, 4.0, 5.4, 6.8)$, whose gaps from the observed y are $(-0.2, 0.4, 0, -0.4, 0.2)$ and sum to zero — as the next section proves they always must.

On the thirty markets, NumPy fits the line in one call, and the two readings of the slope agree to the last digit:

```
import numpy as np

# The least-squares line, straight from a one-line fit: polyfit of degree 1
# returns the slope first, then the intercept -- no matrices, no iteration.
a, b = np.polyfit(x, y, 1)
print("slope a      :", round(a, 4), "k€ sales per extra k€ spend")
print("intercept b:", round(b, 4), "k€")

# The same slope, read off r and the two standard deviations: a = r * sy / sx.
r = np.corrcoef(x, y)[0, 1]
print("r * sy / sx:", round(r * np.std(y, ddof=1) / np.std(x, ddof=1), 4))
```

```
slope a      : 1.8502 k€ sales per extra k€ spend
intercept b: 4.0662 k€
r * sy / sx: 1.8502
```

Figure 2 draws the fitted line through the cloud. The slope $\hat{a} = 1.85$ is the number a manager reads first.

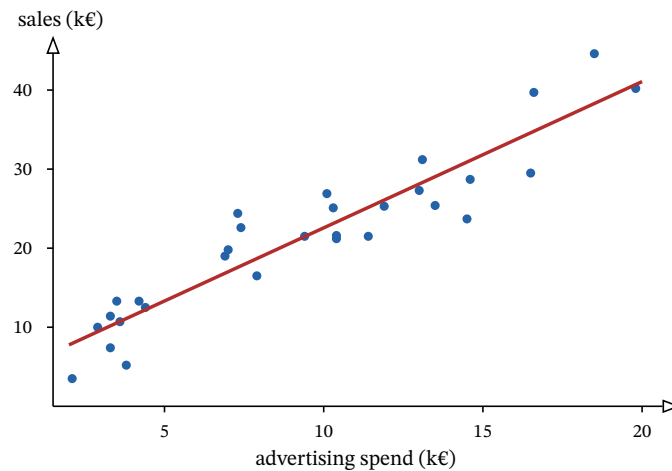


Figure 2: The thirty markets and their least-squares line $\hat{y} = 1.85x + 4.07$. The slope says that each extra thousand euros of weekly spend goes with about 1.85 k€ more sales; the vertical gaps from the points to the line are the residuals, which the fit makes as small as possible in the squared-sum sense.

2.3. The slope as a business quantity

The slope is not only a number on a plot; it is the quantity a decision-maker acts on. Here $\hat{a} \approx 1.85$ means: across these markets, each additional thousand euros of weekly advertising goes with about 1.85 thousand euros of additional sales. Its units — thousands of euros of sales per thousand euros of spend — make it directly comparable with the cost of the spend it describes, which is why a manager reads the slope first.

Pitfall. Read the slope as “goes with”, not “causes”. The slope is descriptive: it summarises how sales have varied with spend across these markets. Reading it as “if we raise spend, sales will rise by 1.85 per unit” assumes that no hidden third variable is at work and that a past pattern would survive a deliberate change — assumptions the data alone cannot support. Chapter 6 shows a slope whose very sign is the artefact of a left-out variable.

2.4. Two identities every residual satisfies

Once the line is fitted, the gaps it leaves have their own name and their own symbol. They are a different object from the model error introduced in Chapter 4, and the difference matters, so we define them carefully now.

Definition 8 (Residual). For the least-squares line, the **fitted value** at market i is $\hat{y}_i = \hat{a}x_i + \hat{b}$, and the **residual** there is

$$e_i = y_i - \hat{y}_i.$$

It is the vertical gap between the observed point and the fitted line, a quantity computed from the data.

The two equations we set to zero in the derivation were not arbitrary: read at the fitted line, each says something about the residuals.

Proposition 3 (Residual identities). The residuals of the least-squares line satisfy

$$\sum_{i=1}^n e_i = 0 \quad \text{and} \quad \sum_{i=1}^n x_i e_i = 0.$$

Proof. Each identity is one of the two equations of the derivation, read at the fitted slope and intercept.

The intercept equation. Step 1 of the derivation was $\sum_i (y_i - \hat{a}x_i - \hat{b}) = 0$; the summand is exactly e_i , so $\sum_i e_i = 0$.

The slope equation. Step 2 was $\sum_i x_i (y_i - \hat{a}x_i - \hat{b}) = 0$; the summand is exactly $x_i e_i$, so $\sum_i x_i e_i = 0$. $\square$

Both identities can be read straight off a residual plot. That $\sum_i e_i = 0$ says the residuals are centred at zero: their positive and negative parts exactly balance, so a plot of the residuals sits evenly above and below the horizontal line at zero. That $\sum_i x_i e_i = 0$ says the residuals carry no leftover linear trend in x : plotted against x , they show no slope of their own, because any such slope has already been absorbed into the fitted line.

We can watch both hold on the real fit; up to the tiny rounding error of floating-point arithmetic, they are zero.

```
import numpy as np

# Two identities the least-squares residuals ALWAYS satisfy, by construction:
# they are exactly the two normal equations, so both hold to machine precision.
a, b = np.polyfit(x, y, 1)
y_hat = a * x + b
e = y - y_hat

print("sum of e_i      :", np.sum(e))
print("sum of x_i * e_i :", np.sum(x * e))
```

```
sum of e_i      : -2.0250467969162855e-13
sum of x_i * e_i : -1.9184653865522705e-12
```

Pitfall. That $\sum_i e_i = 0$ holds by construction — it is true for any data at all, even a cloud with no real trend. So confirming it tells you only that the line was fitted correctly; it is not evidence that the line is a good model. The genuine checks — how much variation the line explains, and whether the residuals behave — come in the next two chapters.

3. Decomposing the variance: R^2

A line has been fitted. How good is it? The natural yardstick asks how much of the variation in sales the line accounts for. Answering it splits the total variation of y into a part the line explains and a part it leaves behind — a split that yields the single most-quoted number in applied statistics, R^2 .

3.1. Three sums of squares

Measure variation, as always, by summed squared deviations. There are three to track: the spread of the observed y about their mean, the spread of the fitted values about that same mean, and the spread of the residuals.

Definition 9 (Total, explained and residual sums of squares). For the least-squares line, with fitted values $\hat{y}_i = \hat{\alpha}x_i + \hat{\beta}$ and residuals $e_i = y_i - \hat{y}_i$, the three **sums of squares** are

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2, \quad \text{ESS} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2, \quad \text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

Here TSS is the **total** sum of squares, all the variation to be explained; ESS is the **explained** sum of squares, the variation the fitted values reproduce; and RSS is the **residual** sum of squares, the variation left in the residuals. The total sum of squares is the S_{yy} of Chapter 1: $\text{TSS} = S_{yy}$.

Pitfall. These three-letter names are not used consistently across textbooks: some write “SSE” for the residual (error) sum of squares — the opposite of the meaning of ESS here. In this book ESS is explained and RSS is residual. Read any other source’s own definitions before trusting its letters. What never changes is the plain-language split: total equals explained plus left over.

3.2. The decomposition and R^2

The three sums are not independent: the explained and residual parts add back exactly to the total.

Theorem 2 (Variance decomposition). For the least-squares line,

$$\text{TSS} = \text{ESS} + \text{RSS}.$$

Proof. We work in two steps.

Step 1 — split each deviation and square. Write each centred observation as its explained part plus its residual,

$$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i) = (\hat{y}_i - \bar{y}) + e_i,$$

then square and sum over i :

$$\text{TSS} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n e_i^2 + 2 \sum_{i=1}^n (\hat{y}_i - \bar{y})e_i = \text{ESS} + \text{RSS} + 2 \sum_{i=1}^n (\hat{y}_i - \bar{y})e_i.$$

Step 2 — kill the cross term. Since $\hat{y}_i = \hat{a}x_i + \hat{b}$ and $\bar{y} = \hat{a}\bar{x} + \hat{b}$, subtracting gives $\hat{y}_i - \bar{y} = \hat{a}(x_i - \bar{x})$. Hence

$$\sum_{i=1}^n (\hat{y}_i - \bar{y})e_i = \hat{a} \sum_{i=1}^n (x_i - \bar{x})e_i = \hat{a} \left(\sum_{i=1}^n x_i e_i - \bar{x} \sum_{i=1}^n e_i \right) = 0,$$

because both residual identities of Chapter 2 make the two inner sums zero. The cross term vanishes, leaving $\text{TSS} = \text{ESS} + \text{RSS}$. $\square$

The decomposition turns the quality of the fit into a single ratio.

Definition 10 (Coefficient of determination R^2). For the least-squares line, the **coefficient of determination** is the share of the total variation that the line explains,

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}.$$

Proposition 4 (R^2 lies in $[0, 1]$, and $R^2 = r^2$). For the least-squares line, $0 \leq R^2 \leq 1$, and moreover $R^2 = r^2$: the share of variation explained is exactly the square of the correlation.

Proof. We work in three steps.

Step 1 — divide the decomposition by the total. Since $\text{TSS} > 0$, dividing $\text{TSS} = \text{ESS} + \text{RSS}$ by TSS gives

$$1 = \frac{\text{ESS}}{\text{TSS}} + \frac{\text{RSS}}{\text{TSS}}.$$

Both ESS and RSS are sums of squares, so both are ≥ 0 ; each ratio therefore lies in $[0, 1]$, and in particular $R^2 = \text{ESS} / \text{TSS} \in [0, 1]$.

Step 2 — an explicit formula for the explained sum of squares. Using $\hat{y}_i - \bar{y} = \hat{a}(x_i - \bar{x})$ again,

$$\text{ESS} = \sum_{i=1}^n \hat{a}^2 (x_i - \bar{x})^2 = \hat{a}^2 S_{xx} = (S_{xy}/S_{xx})^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}}.$$

Step 3 — conclude. With $\text{TSS} = S_{yy}$,

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = \frac{S_{xy}^2}{S_{xx}S_{yy}} = \left(S_{xy} / \sqrt{S_{xx}S_{yy}} \right)^2 = r^2.$$

$\square$

The identity $R^2 = r^2$ is worth pausing on. An r of 0.93 is an R^2 of about 0.87: the line accounts for roughly 87% of the variation in sales.

Example — reading R^2 off a small fit. For the five-market line of Chapter 2, $\bar{y} = 4$ and the fitted values are (1.2, 2.6, 4, 5.4, 6.8). Then $\text{ESS} = \sum_i (\hat{y}_i - 4)^2 = 7.84 + 1.96 + 0 + 1.96 + 7.84 = 19.6$ and

$RSS = \sum_i e_i^2 = 0.04 + 0.16 + 0 + 0.16 + 0.04 = 0.4$, so $TSS = 20$ and $R^2 = 19.6/20 = 0.98$. The line explains 98% of the variation — and indeed $r = 0.99$, so $r^2 = 0.98$.

On the thirty markets the three sums, and their ratio, come out as follows.

```
import numpy as np

# Split the total variation of y into the part the line reproduces and the part
# left in the residuals: TSS = ESS + RSS, and R-squared is the explained share.
a, b = np.polyfit(x, y, 1)
y_hat = a * x + b

TSS = np.sum((y - np.mean(y)) ** 2)      # total      : spread of y about its mean
ESS = np.sum((y_hat - np.mean(y)) ** 2)   # explained: spread of the fitted values
RSS = np.sum((y - y_hat) ** 2)            # residual : spread left in the residuals

print("TSS (total)      :", round(TSS, 1))
print("ESS (explained):", round(ESS, 1))
print("RSS (residual) :", round(RSS, 1))
print("ESS + RSS       :", round(ESS + RSS, 1))
print("R2 = ESS / TSS :", round(ESS / TSS, 4))
print("r squared       :", round(np.corrcoef(x, y)[0, 1] ** 2, 4))
```

```
TSS (total)      : 2933.3
ESS (explained): 2559.7
RSS (residual)   : 373.6
ESS + RSS        : 2933.3
R2 = ESS / TSS   : 0.8726
r squared        : 0.8726
```

The printout confirms all three facts at once: $ESS + RSS = TSS$ to the decimal, $R^2 = 0.8726$, and $r^2 = 0.8726$. Laid out as a table, the same three sums form the variance-decomposition table that regression software prints, with R^2 read straight off the “share of total” column.

Source of variation	Sum of squares	Share of total
Explained by the line (ESS)	2559.7	0.873
Residual, left over (RSS)	373.6	0.127
Total (TSS)	2933.3	1.000

Table 1: The variance-decomposition table for the thirty markets. The total variation splits into the part the line explains and the part left in the residuals; $R^2 = ESS / TSS = 0.8726$ is the explained row’s share of the total — the number quoted as “the model explains 87% of the variation”.

Figure 3 shows where the decomposition comes from, one observation at a time.

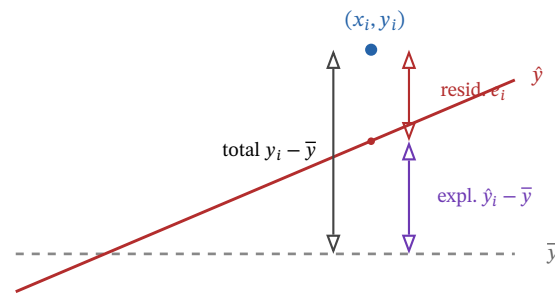


Figure 3: The decomposition, one point at a time. The gap from a point to the mean line $\bar{y}$ (total) splits into the gap from the fit to the mean (explained) plus the gap from the point to the fit (residual). Squared and summed over all points — with the cross terms cancelling by the residual identities — this is $TSS = ESS + RSS$, and R^2 is the explained share.

Pitfall. R^2 measures linear fit only, and it says nothing about whether the model is appropriate. The four famous Anscombe datasets share $R^2 = 0.67$ while looking utterly different — one a clean line, one a parabola, one ruined by a single outlier. A high R^2 earns a look at the residuals, not the end of the analysis. That look is the next chapter.

4. Residuals and the normality assumption

R^2 says how much the line explains; it does not say whether the line is trustworthy. Trust rests on assumptions about the errors, and the residuals e_i are our only window onto them. This chapter states the assumptions, then reads the diagnostic plots that check them.

4.1. The inferential model

So far the line has been pure description. To test it — the business of the next chapter — we must say what the scatter around the line is like. The standard assumption promotes the fitted line to a model with a genuine random error.

Definition 11 (Simple linear regression model). The **simple linear regression model** states that the observations are generated as

$$y_i = ax_i + b + \varepsilon_i, \quad \text{with } \varepsilon_1, \dots, \varepsilon_n \text{ independent and each } \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

Here a and b are the unknown **true** slope and intercept, and ε_i is the **error** of market i . The errors are **centred** ($E(\varepsilon_i) = 0$, so the line is right on average), have a **common variance** σ^2 , are **independent**, and are **normally distributed**.

Two objects must not be confused. The error ε_i belongs to the model: it is the true, unobservable deviation of market i from the true line $ax + b$. The residual $e_i = y_i - \hat{y}_i$ of Chapter 2 belongs to the data: it is the observable gap from the fitted line $\hat{a}x + \hat{b}$. We never see the errors; the residuals are our best proxy for them, and every diagnostic in this chapter reads the errors' behaviour off the residuals.

The same distinction applies to the slope. The true slope a is a fixed, unknown number in the model; the estimate $\hat{a} = S_{xy}/S_{xx}$ is a number computed from the sample. A different thirty markets would give a different $\hat{a}$, but the same a . Keeping the two apart is what makes the next chapter's test meaningful.

Why assume the errors are normal? Each error is the combined effect of many small, unrelated influences the line leaves out — local weather, a competitor's promotion, a data-entry slip — and, by the Central Limit Theorem you met earlier, the sum of many small independent effects is approximately normal. Normality is not sacred, but it is the assumption under which the slope's t -test of the next chapter is exactly valid; when it fails badly, that test becomes an approximation at best.

4.2. Reading the residuals

Checking the assumptions is a visual skill. Three plots answer three questions, and all three are read on the residuals of our thirty-market fit.

Are the errors normal? A histogram of the residuals should look roughly like a symmetric mound centred at zero. Figure 4 shows the thirty residuals: centred near zero, with no long tail on either side, but distinctly lumpy — two separate peaks straddling zero rather than one clean central mound. That lumpiness is the point to absorb here. Thirty points make a histogram coarse: a sample drawn from a genuinely normal error is routinely uneven, even mildly two-humped, from sampling variation alone. A histogram this small can therefore neither confirm normality nor rule it out — only a gross

departure, such as a long one-sided tail, would show through the noise. For a sharper read we turn to the QQ-plot.

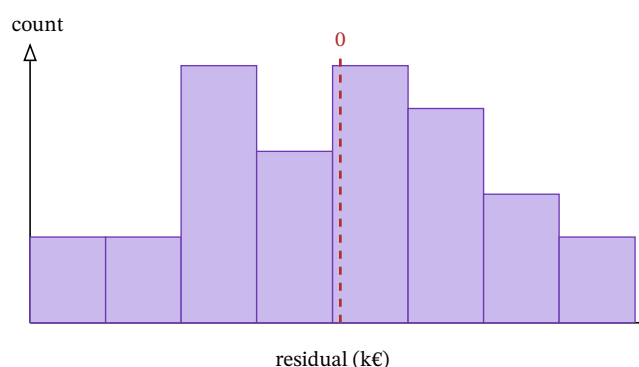


Figure 4: Histogram of the thirty residuals. It is roughly symmetric and centred at zero (dashed), with no heavy tail, but lumpy — its two peaks straddling zero are the kind of unevenness sampling variation alone produces at thirty points. A histogram this small can neither confirm nor rule out normality; only a gross departure, such as a long one-sided tail, would show through the noise.

Are the errors normal — the sharper test? A **quantile–quantile plot**, or QQ-plot, reads more from the same thirty numbers than a histogram can. It sorts the residuals, divides them by their standard deviation so that they share the scale of a standard normal, and plots these **standardised residuals** against the values a standard normal sample of the same size would give. If the residuals are normal, the points fall along the line $y = x$, the 45-degree line, because each standardised residual then matches its normal quantile. Figure 5 shows ours close to that line, with only the mild wandering in the tails that thirty points always show. A QQ-plot is a picture to read by eye, not a computation to run.

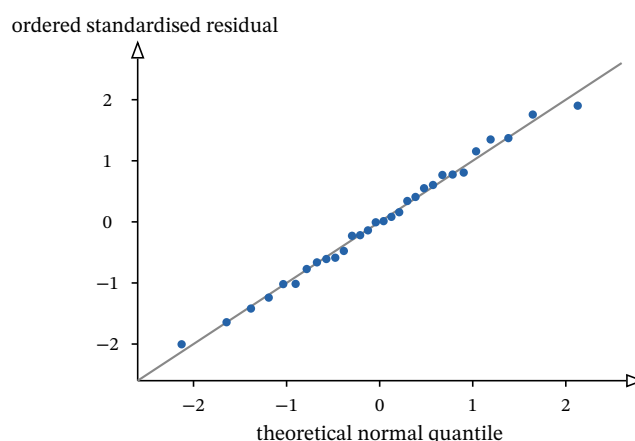


Figure 5: Normal QQ-plot of the residuals. A point on the grey 45-degree line means the residual matches its normal quantile. These track the line closely, bending only slightly in the tails, so the normal assumption is reasonable here.

Is the variance constant? Plot the residuals against the fitted values. Under the model the band of residuals should have constant width — the same scatter everywhere, with no pattern. The two words for this are worth fixing.

Definition 12 (Homoscedasticity and heteroscedasticity). The errors are **homoscedastic** when they share one common variance σ^2 , so the scatter of the points about the line is the same

at every value of x . They are **heteroscedastic** when that variance changes with x , so the scatter widens or narrows across the range.

On a plot of residuals against fitted values, homoscedasticity shows as a band of even width and heteroscedasticity as a widening or narrowing one.

Figure 6 puts our fit's residuals (left) beside a second, constructed market (right) whose errors grow with spend — the characteristic widening, or **funnel**, of heteroscedasticity.

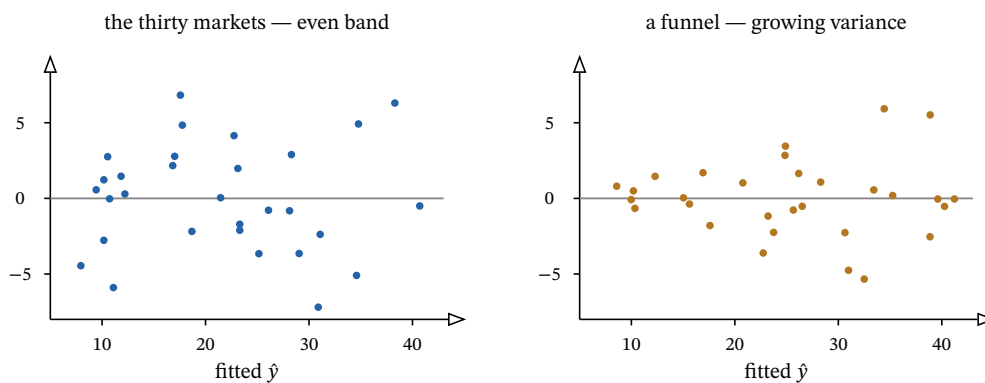


Figure 6: Residuals against fitted values. Left: the thirty markets' own residuals, an even band with no clear widening — the equal-variance assumption is reasonable. Right: a constructed market whose scatter fans out as the fitted value grows, an unmistakable funnel. Under a funnel the fitted line stays right on average, but it is no longer the most precise line, and an error bar built on a single common σ^2 understates the uncertainty where the data are noisy.

Reading the width of the band by eye is the intended skill, but we can put one number on it. We recompute the residuals of our fit and — since the markets are listed in order of increasing spend, and so of increasing fitted value — compare the residual standard deviation in the first ten markets, the lowest-fitted third, with the last ten, the highest.

```
import numpy as np

# Residuals of the thirty-market fit, and how their spread changes across the
# fitted range. The markets are listed in increasing order of spend, and the
# slope is positive, so the fitted values increase down the list: the first ten
# markets are the lowest-fitted third and the last ten the highest. Compare the
# residual standard deviation in each -- a roughly constant spread is what one
# common variance looks like on the residual plot.
a, b = np.polyfit(x, y, 1)
y_hat = a * x + b
e = y - y_hat

print("residual sd, low-fitted third :", round(np.std(e[:10], ddof=1), 2), "k€")
print("residual sd, high-fitted third:", round(np.std(e[-10:], ddof=1), 2), "k€")
```

```
residual sd, low-fitted third : 2.91 k€
residual sd, high-fitted third: 4.3 k€
```

The residual standard deviation across the fitted range rises only mildly, from about 2.91 k€ in the lowest-fitted third to about 4.3 k€ in the highest. For groups of ten that difference is well within what sampling alone produces, so we read the left panel as an even band and judge the equal-variance assumption reasonable here.

Remark. The usual remedies: for a funnel, transform the response (a logarithm often steadies a widening spread) or give the noisy points less weight; for a curved residual pattern — residuals positive at both ends and negative in the middle — the straight line is the wrong shape, and the fix is to enrich the model, for instance by adding a squared term so that it bends while staying linear in its coefficients.

5. Testing the slope

Our estimate is $\hat{a} = 1.85$. But a slope computed from thirty noisy markets is itself a random quantity: a different thirty markets would give a different number. Could the true slope a of the model be zero — no linear link between spend and sales — with $\hat{a} = 1.85$ arising from noise alone? That question is a hypothesis test, and it is the question a regression summary exists to answer. The test is about the model's true slope a , not about the estimate $\hat{a}$: we ask whether the data are consistent with $a = 0$.

5.1. The sampling distribution of the slope

Under the model of Chapter 4 the estimate $\hat{a}$ is a random variable, and we can pin down its mean and variance exactly.

Proposition 5 (Mean and variance of the slope). Under the model $y_i = ax_i + b + \varepsilon_i$, with the ε_i independent of mean zero and variance σ^2 and the x_i treated as fixed,

$$\mathbb{E}(\hat{a}) = a, \quad \mathbb{V}(\hat{a}) = \frac{\sigma^2}{S_{xx}}.$$

The estimate is unbiased, and its spread shrinks as the x -values spread out (larger S_{xx}) or the noise shrinks (smaller σ^2).

Proof. Write the slope as a weighted sum of the responses. Since $\sum_i (x_i - \bar{x}) = 0$, the $\bar{y}$ term drops out of the numerator, giving

$$\hat{a} = \frac{S_{xy}}{S_{xx}} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}} = \sum_{i=1}^n c_i y_i, \quad c_i = \frac{x_i - \bar{x}}{S_{xx}}.$$

The weights satisfy $\sum_i c_i = 0$ and $\sum_i c_i x_i = 1$. Hence, by linearity of expectation,

$$\mathbb{E}(\hat{a}) = \sum_i c_i (ax_i + b) = a \sum_i c_i x_i + b \sum_i c_i = a,$$

and, since the y_i are independent with variance σ^2 ,

$$\mathbb{V}(\hat{a}) = \sum_i c_i^2 \mathbb{V}(y_i) = \sigma^2 \sum_i c_i^2 = \sigma^2 \frac{\sum_i (x_i - \bar{x})^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}}.$$

□

We do not know σ^2 , so we estimate it from the residuals. Here the divisor is $n - 2$, not n or $n - 1$: the residuals are gaps from the fitted line, and that line already spent two parameters — the slope $\hat{a}$ and the intercept $\hat{b}$ — both estimated from the same data. Two of the n residuals are therefore determined by the other $n - 2$ (that is what the two residual identities of Chapter 2 say), so only $n - 2$ of them are free. Dividing by the number of free residuals makes the estimate unbiased.

$$s^2 = \frac{\text{RSS}}{n - 2}, \quad \text{SE}(\hat{a}) = \frac{s}{\sqrt{S_{xx}}}.$$

The quantity $SE(\hat{a})$ is the **standard error of the slope**: the estimated standard deviation of $\hat{a}$.

5.2. The t -statistic and the p -value

To test $H_0 : a = 0$ against $H_1 : a \neq 0$, measure how many standard errors the estimate sits away from zero.

Framework 1 (The slope t -test). For the simple linear regression model, under $H_0 : a = 0$ the statistic

$$t = \frac{\hat{a}}{SE(\hat{a})} = \frac{\hat{a}\sqrt{S_{xx}}}{s} \quad \text{follows the Student law } t_{n-2}.$$

The two-sided p -value is $\mathbb{P}(|T| \geq |t|)$ for $T \sim t_{n-2}$: the probability, if the true slope were zero, of a slope estimate at least this many standard errors from zero. Reject H_0 at level α (usually 0.05) when $|t|$ exceeds the critical value $t_{n-2}^{1-\alpha/2}$ read from the t -table, equivalently when $p < \alpha$.

Why the Student law t_{n-2} , and not the normal? Because we replaced the unknown σ by the estimate s ; the extra uncertainty in that estimate fattens the tails into a Student law, and its degrees of freedom are the $n - 2$ free residuals that s is built from — the same $n - 2$ that appeared in its divisor. We assemble the test from the residual sum of squares, and compare the statistic by hand with the tabulated critical value.

```
import numpy as np

# Could a slope this steep come from noise alone? Test H0: a = 0 against
# H1: a != 0. Estimate the noise level, form the standard error of the slope,
# read off the t statistic -- then compare it BY HAND to the t-table value.
a, b = np.polyfit(x, y, 1)
y_hat = a * x + b
e = y - y_hat

RSS = np.sum(e ** 2)
se = np.sqrt(RSS / (n - 2))          # residual standard error
Sxx = np.sum((x - np.mean(x)) ** 2)
se_a = se / np.sqrt(Sxx)           # standard error of the slope
t = a / se_a

print("slope a          :", round(a, 4))
print("std error se_a   :", round(se_a, 4))
print("t = a / se_a     :", round(t, 2), " (df = n - 2 =", n - 2, ")")
print("t-table 5% value :", 2.048)
```

```
slope a          : 1.8502
std error se_a   : 0.1336
t = a / se_a     : 13.85 (df = n - 2 = 28 )
t-table 5% value : 2.048
```

The slope sits $t = 13.85$ standard errors from zero, on $n - 2 = 28$ degrees of freedom. The two-sided 5% critical value from the t -table is $t_{28}^{0.975} = 2.048$; our $|t| = 13.85$ is far larger than it, so we reject H_0 .

decisively. Figure 7 shows how extreme this is: the whole t_{28} distribution lives between about ± 3 , so a value near 14 falls far outside the range the plot can show.

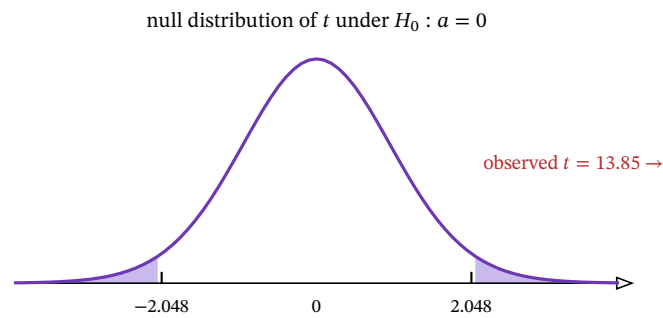


Figure 7: The null distribution t_{28} . If the true slope were zero, the t -statistic would scatter within about ± 3 ; the shaded tails beyond the 5% critical values ± 2.048 are the rejection region. Our observed $t = 13.85$ lies far to the right of everything shown, so the p -value is minute.

A regression package would print exactly the numbers we assembled, together with the p -value it computes from the same t_{28} law. Table 2 shows that software summary for our fit; reading such a table — slope, standard error, t -statistic, p -value — is the everyday form of this test.

Quantity	Value
Slope $\hat{a}$	1.8502
Standard error $SE(\hat{a})$	0.1336
t -statistic = $\hat{a}/SE(\hat{a})$	13.85
p -value (two-sided, t_{28})	4.7e-14
R^2	0.8726

Table 2: The regression summary a package prints for the thirty-market fit. The p -value 4.7e-14 is the chance, were the true slope zero, of a t -statistic as extreme as 13.85 it is far below the usual 0.05 threshold, so the slope is **statistically significant** — unlikely to be an accident of noise.

Pitfall. A p -value is not the probability that H_0 is true, and statistical significance is not practical importance. With enough data a tiny slope becomes significant; with few data a large one may not. Significance says only that the sign and the existence of the effect are unlikely to be noise. Whether the effect is large enough to act on is a separate question, read from the size of $\hat{a}$ itself in its business units.

6. Simpson's paradox and confounding

Every slope so far has been read as a quantity to act on: raise x , expect $\hat{a}$ more units of y . This chapter shows how that reading can be not merely imprecise but backwards — the fitted slope pointing the opposite way to the real, within-group relationship — when a confounding variable is left out. This is **Simpson's paradox**, the sharpest form of “correlation is not causation”. Recall from Chapter 1 that a confounder is a third variable influencing both x and y .

6.1. A regression slope that changes sign

Consider a firm selling products across three quality tiers — budget, mid, premium. For each product we record its price and its weekly quantity sold. Economic sense says that, holding quality fixed, a higher price should mean a lower quantity: a negative price–quantity slope. Fit the pooled data, ignoring the tier, and the slope comes out positive.

```
import numpy as np

# Fifteen products across three quality tiers. Within every tier, a higher price
# goes with a lower quantity sold. Watch the POOLED slope once the tier -- the
# confounder -- is ignored and all fifteen products are fitted together.
price = np.array([10, 11, 12, 13, 14, 16, 17, 18, 19, 20, 24, 25, 26, 27, 28])
qty = np.array([64, 62, 63, 60, 61, 92, 90, 91, 88, 89, 120, 118, 119, 116, 117])
tier = np.array(["budget"] * 5 + ["mid"] * 5 + ["premium"] * 5)

a_all, b_all = np.polyfit(price, qty, 1)
print("pooled slope, all 15 products:", round(a_all, 2))
for name in ["budget", "mid", "premium"]:
    m = (tier == name)
    a_g, b_g = np.polyfit(price[m], qty[m], 1)
    print("slope within " + name + " tier:", round(a_g, 2))
```

```
pooled slope, all 15 products: 3.7
slope within budget tier: -0.8
slope within mid tier: -0.8
slope within premium tier: -0.8
```

Within every tier the slope is -0.8 , yet the pooled slope is $+3.7$. Figure 8 shows why.

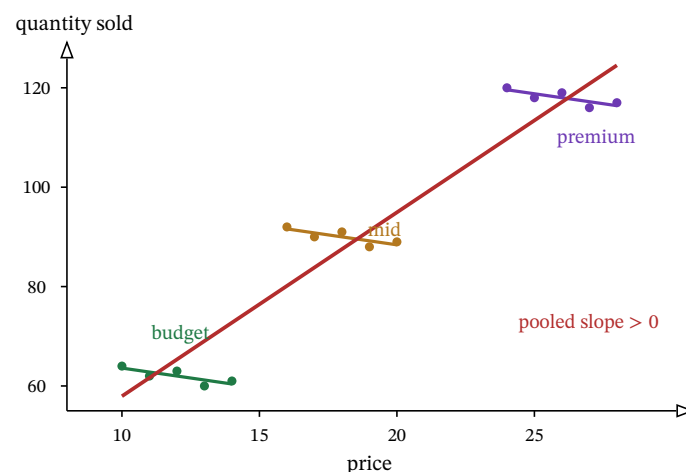


Figure 8: Simpson's paradox in a regression slope. Within each quality tier the price–quantity slope is negative (three coloured lines): dearer items sell less. But premium tiers sit up and to the right — higher price and higher quantity, because quality lifts both — so the single pooled line (red) slopes upward. The confounder is quality; omit it and the slope reverses.

The pooled slope blends two different comparisons: the within-tier effect (move along one coloured line) and the between-tier effect (step from one tier's cloud to the next). Quality raises price and raises quantity at the same time, so the between-tier steps rise from left to right, a steep positive trend. Stack the tiers into one undifferentiated cloud and that between-tier climb dominates the gentle within-tier declines, so the fitted line takes its sign. Quality is confounded with price: it moves with price and it moves quantity, so its effect is silently attributed to price.

6.2. The same reversal in a table of rates

The paradox is not special to regression slopes; it appears wherever groups are pooled. A classic case compares two treatments for kidney stones by their success rates. Table 3 records, for each treatment, the successes out of the patients treated, split by whether the stone was small or large.

	Small stones	Large stones	Both, pooled
Treatment A	81/87 = 93%	192/263 = 73%	273/350 = 78%
Treatment B	234/270 = 87%	55/80 = 69%	289/350 = 83%

Table 3: Success rates of two kidney-stone treatments, from Charig and colleagues (1986). Treatment A does better on small stones (93% against 87%) and better on large stones (73% against 69%), yet worse overall (78% against 83%). The confounder is stone size: the harder large stones were given to Treatment A far more often, dragging its pooled rate down.

Read across the first two columns and Treatment A wins in both: 93% against 87% on small stones, and 73% against 69% on large stones. Read the pooled column and Treatment A loses: 78% against 83%. The reversal comes from the same mechanism as the regression slope. Stone size is a confounder: large stones are both harder to treat successfully and were assigned to Treatment A far more often (263 of its 350 cases, against 80 of Treatment B's). Treatment A's pooled rate is pulled down by the hard cases it was handed, not by any weakness of the treatment. This is the same paradox as the price example, now on counts and rates rather than on a fitted line.

6.3. Why more data does not help, and what does

Pitfall. The paradox is not cured by “collecting more data”. Every tier, or every stone size, could hold thousands of cases and the reversal would stand. It is cured only by **conditioning** on the confounder: analyse within each group — as the code does by tier, and as the table does by stone size — or bring the confounder into the model so that its effect is estimated rather than charged to x . The remedy is a better model, not a bigger sample.

Remark. Which slope is “right” depends on the question. To advise a manager how a single product’s sales respond to its own price, the within-tier slope is the quantity to use. The pooled positive slope answers a different, descriptive question — “do more expensive products sell more?” — and yes, they do, because the expensive ones are of higher quality. The mistake is to read the second number as an answer to the first.

Primary sources and further reading

The methods in this book are older than the software that runs them, and each traces back to a readable original.

- **Correlation.** K. Pearson (1896), “Mathematical Contributions to the Theory of Evolution. III. Regression, Heredity, and Panmixia,” *Philosophical Transactions of the Royal Society A* — the paper that fixed the coefficient r and the word “regression” in their modern senses.
- **Simpson’s paradox.** E. H. Simpson (1951), “The Interpretation of Interaction in Contingency Tables,” *Journal of the Royal Statistical Society B* — the short note behind Chapter 6.
- **The kidney-stone data.** C. R. Charig, D. R. Webb, S. R. Payne and J. E. Wickham (1986), “Comparison of treatment of renal calculi by open surgery, percutaneous nephrolithotomy, and extracorporeal shockwave lithotripsy,” *British Medical Journal* — the source of Table 3.
- **A modern applied treatment.** G. James, D. Witten, T. Hastie and R. Tibshirani, *An Introduction to Statistical Learning* (Springer) — Chapter 3 develops simple and multiple linear regression, the slope’s test, and residual diagnostics without matrix prerequisites.
- **A gentle inferential companion.** D. Diez, M. Çetinkaya-Rundel and C. Barr, *OpenIntro Statistics* (free online) — thorough on standard errors, t -tests, and the interpretation of R^2 .